HOANG CAO PHONG

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EDUCATION

Vietnam National University, Ho Chi Minh City – University of Science (HCMUS) **Sep 2024 – Ongoing**
Bachelor of Science | English Intensive Program in Information Technology

- **Specialization:** Computer Science – Computer Vision track
- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Linear Algebra, Probability and Statistics, Calculus, General Physics, Software Engineering, Database, Machine Learning, Computer Vision, Digital Image and Video Processing, Natural Language Processing, Deep Learning, Parallel Programming.
- ***Honor:*** *Selected for University ICPC Team – University of Science, VNU-HCM (2024)*

Tran Hung Dao High School for the Gifted, Phan Thiet City, Binh Thuan, Vietnam **Sep 2021 – Jun 2024**

- ***Honor:*** Third Prize, Provincial Informatics Olympiad

AWARDS & CERTIFICATIONS

- HPC Summer School – HCMUT 2026 - Certificate of Completion **Jun 2026**
- Quantum Computing and Your Future Course - Certificate of Completion, Physics@VNUHCM-US **Jun 2026**
- Outstanding Trainer Award - Gemini Academy for Student, Google **Jun 2026**
- Applying AI in Learning and Research - Issued by Student Culture House of Ho Chi Minh City **Apr 2025**
- Medical Imaging workshop - Certified by Kyoto Institute of Technology, Japan **Feb 2025**
- Robot Programming - Certified by United States of America Consulate General – Ho Chi Minh City **Oct 2024**
- Third Prize - Provincial Informatics Olympiad **Oct 2023**

RESEARCH EXPERIENCE

UIT Information Security Lab (UIT - InSecLab) | AI & Security Research Intern **Mar 2026 – Ongoing**

- Co-authored (as second author) an accepted full paper at IEEE ICPADS 2026, Tokyo, Japan on small language models for smart-contract vulnerability detection.
- Led a small research group on AI-assisted cybersecurity, focusing on security analysis and multimodal AI.
- Supported research workflows, including literature review, data preprocessing, model evaluation, and result analysis.
- Assisted in developing and maintaining web-based resources for the lab and CNSC Information Security Center.

PUBLICATIONS & APPEARANCE

The 32nd IEEE International Conference on Parallel and Distributed Systems **IEEE ICPADS 2026**
Accepted Full Paper, Tokyo, Japan, 2026 | CORE Rank B **Presenting Author**

- *"Small Language Models for Smart Contract Vulnerability Detection: A Fine-Grained Analysis on Capabilities, Localization, and Failure Modes."*
- Doan Minh Trung, **Hoang Cao Phong**, Luc Vinh Kiet, Ho Minh Tri, Nguyen Nhat Dong, Phan The Duy, Tu-Anh Nguyen-Hoang, and Van-Hau Pham

Abstract (InSecLab)

TECHNICAL EXPERIENCE & ACTIVITIES

Google Student Ambassador Program | Google Student Ambassador 2026 **Jan 2026 – Ongoing**

- Spearheaded campus initiatives to advocate for Google's AI and developer ecosystem.
- Orchestrated technical workshops equipping students with practical insights into AI trends.
- Cultivated a dynamic tech community to accelerate cutting-edge technology adoption.

Google Developer Group on Campus – HCMUS | MC & Graphic Designer **Oct 2024 – Ongoing**

- Served as MC for technical workshops and community events, introducing sessions and coordinating program flow.
- Designed visuals for events, workshops, and social media using Illustrator, Photoshop, GIMP, and Premiere.
- Collaborated with developers and organizers to align technical presentations with branding and audience engagement.

Library of University of Science – VNUHCM | Member of Technical Department **Oct 2024 – Ongoing**

- Set up audiovisual systems and online streaming for library events.
- Produced videos, newsletters, and social media updates to boost student engagement.
- Conducted data analysis and research to improve library services.

Strong PC – Student Laptop Service & Optimization | Co-founder, Management Board **Sep 2025 – Ongoing**

- Co-founded and led a student-run laptop support project providing maintenance and optimization services.
- Managed operations, task assignments, and financial planning across technical and communication teams.
- Directed internal documentation, workflow development, and collaboration planning with the University Library

TECHNICAL PROJECTS

Breast Cancer Classification | Tech Lead / ML Engineer **Jun 2026 - Aug 2026**

- **Led a team of five** to deliver an interactive **ML demo** through code reviews and **FastAPI-React** integration.
- Contributed to the team's **93.86% accuracy on 114 held-out samples** and **16.7% fewer leaves** versus baseline through controlled model evaluation and integration.
- Enabled interpretable classification across **30 features** by implementing Decision Trees from scratch with **Gini/Entropy** splitting and class probabilities.
- Evaluated depth–performance trade-offs on **569 samples** using stratified **5-fold cross-validation** to compare custom and scikit-learn trees.

Link Project: *github.com/HoangCaoPhong/breast-cancer-machine-learning*
Website: *bc-ml.zapto.org*

WindyAI – Smart Travel Planner | Project Manager & AI Engineer **Sep 2025 - Dec 2025**

- **Led a team of five** in designing an **end-to-end AI decision support system**, integrating **auxiliary AI modules** *(weather-aware adjustment, LLM-based assistant)*, and exploring reinforcement learning–based planning.
- Formulated **travel itinerary planning** as a **constrained optimization problem** *(time, budget, spatial, preference constraints)*, and designed a **Greedy + Lookahead heuristic** with a **multi-factor POI scoring function** to generate feasible, near-optimal travel sequences under real-world constraints.
- Developed an **image-based landmark recognition module** using a **ResNet-based CNN (PyTorch)** on user-uploaded photos, and integrated visual predictions into the itinerary planning pipeline to reduce ambiguity in POI selection.
- Built an **end-to-end computer vision inference pipeline** *(preprocessing, model inference, top-K prediction with confidence scores)* and tightly coupled recognition outputs with downstream planning decisions.

***Link Project*:** *github.com/HoangCaoPhong/WindyAI-Smart_Travel_Planner*

Vietnamese Facial Emotion Recognition | Computer Vision, Deep Learning **Dec 2025 - Ongoing**

- Developed a **real-time facial emotion recognition system** using **CNNs and OpenCV**, and formulated the task as an evaluation of cross-cultural generalization in facial expression recognition.
- Analyzed **dataset bias** of **FER2013 (Western-centric)** on Vietnamese facial data, identifying performance degradation and misclassification patterns across emotion categories.
- Designed and implemented a **transfer learning and fine-tuning pipeline** with Vietnamese facial images, applying **layer freezing** and **low learning-rate optimization** to adapt pre-trained models under limited data conditions.
- Evaluated model performance using **accuracy and confusion matrices**, and analyzed **limitations related to dataset size and cultural expression variance**, highlighting challenges in emotion recognition across cultural contexts.

***Link Project*:** *github.com/HoangCaoPhong/realtime-facial-emotion-recognition-vietnamese-faces*

MindMeetingAI – Intelligent Meeting Assistant | AI Engineer, NLP & LLM Systems **Sep 2025 – Dec 2025**

- Designed an **end-to-end NLP system** for meeting speech transcription, structured summarization, and semantic question answering, framing the task as an applied study of **information retrieval and generation over speech data**.
- Implemented **Vietnamese ASR** using **Whisper**, and built a **transcript normalization pipeline** to improve downstream retrieval and summarization quality.
- Designed a **Retrieval-Augmented Generation (RAG) pipeline** using **sentence embeddings** and a **vector database**, enabling context-aware semantic search over long meeting transcripts.
- Integrated **local LLM inference** for privacy-preserving analysis, and analyzed **RAG effectiveness and failure cases** in summarization and semantic Q&A over noisy ASR outputs.

Keywords: *Speech-to-Text · RAG · LLM Systems · Semantic Search · Sentence Transformers · Vector Databases*
***Link Project*:** *github.com/HoangCaoPhong/MindMeetingAI*

SKILLS

Programming Languages: Python, C/C++, LaTeX, MATLAB, C#, HTML, JavaScript, CSS.
ML/DL: PyTorch, CNNs, Transformers, Transfer Learning, Model Evaluation & Analysis.
CV/NLP: OpenCV, FER, ASR, Embeddings, RAG
Frameworks/Tools: Git/Github, Overleaf, Unity, Notion, Google Workspace, Microsoft Office, WordPress.
Soft Skills: Leadership, Teamwork, Problem Solving, Communication, Time Management, Self-directed learning
Languages: English (fluent), Vietnamese (native speaker), Chinese (beginner).